

Luke Wiley

Q1.2.1) DFS: (3,3,0,0,L) (2,2,1,1,R) (3,2,0,1,L) (3,0,0,3,R) (3,1,0,2,L) (1,1,2,2,R)
(2,2,1,1,L)(0,2,3,1,R) (0,3,3,0,L) (0,1,3,2,R) (0,2,3,1,L) (0,0,3,3,R)
Node Expansions: 13

BFS: (3,3,0,0,L) (3,1,0,2,R) (3,2,0,1,L) (3,0,0,3,R) (3,1,0,2,L) (1,1,2,2,R)
(2,2,1,1,L) (0,2,3,1,R) (0,3,3,0,L) (0,1,3,2,R) (1,1,2,2,L) (0,0,3,3,R)
Node Expansions: 15

Q1.2.2) BFS and DFS behave differently in the way they search due to DFS utilizing a Stack, and BFS utilizing a queue to perform the search. This means that DFS will follow its chosen path as deep as it goes, whether that be a goal state or a dead end. If it reaches the goal state, it will return this path. If not, it will move up a node and perform a search from all child nodes not already explored here, going as deep as it can first before moving to the next one. BFS will search all nodes at the current level before moving to the next one using a queue to do this. Both algorithms will find a solution if it exists, but because they do it differently, they have tradeoffs. DFS may explore less nodes, but may not return the most optimal path, emphasizing efficiency. BFS however may explore more nodes, and is guaranteed to return the optimal solution, at the price of efficiency. This can be seen in the example as BFS explored 15 nodes, but DFS explored 13.

Q2.2.1) UCS Model A: (3,3,0,0,L) (3,1,0,2,R) (3,2,0,1,L) (3,0,0,3,R) (3,1,0,2,L)
(1,1,2,2,R) (2,2,1,1,L) (0,2,3,1,R) (0,3,3,0,L) (0,1,3,2,R)
(0,2,3,1,L) (0,0,3,3,R)
Total Cost: 21
Node Expansions: 15

UCS Model B: (3,3,0,0,L) (3,1,0,2,R) (3,2,0,1,L) (3,0,0,3,R) (3,1,0,2,L)
(1,1,2,2,R) (2,2,1,1,L) (0,2,3,1,R) (0,3,3,0,L) (0,1,3,2,R)
(1,1,2,2,L) (0,0,3,3,R)
Total Cost: 17
Node Expansions: 15

Q2.2.2) In the implementation for UCS, we introduced a new variable in varied costs of certain actions. This can be seen in model A as a missionary move costs 2 to the cannibals 1, and in model B as a move to the right costs 2 and to the left 1. This is different to BFS as the only thing it cared about was the least number of nodes in the solution path possible, where UCS looks for the least total cost of moves based on the weights of certain actions. Even though BFS and both UCS models expanded the same number of nodes, they all took slightly different paths to get to the goal node. Even though both UCS models expanded 15 nodes, their total costs differed, highlighting the difference their cost models made on the paths they took.

Q2.2.3) When the search task has non-uniform costs, it is best to go with UCS as it finds the least cost solution, which would be optimal for the task. If the costs are Uniform, BFS will now also return the least cost solution, which would be optimal for the task because least cost and fewest steps are one in the same. If BFS is used with non-uniform costs, it will return the path with the fewest steps, not necessarily the least cost path, thus a non-optimal solution.

Q3.2.A) Heuristic 1 is admissible. This heuristic takes the total cost of the missionaries and cannibals from the left bank to get to the right bank as its estimate. It is basically getting rid of the rule that the boat can only take two passengers at a time and speculating the cost based on this. This is admissible because at every step it either gives the actual cost to the goal state as its estimate or a an estimate that would cost less than the actual solution (the boat would have to make more moves to move all the people that the heuristic is saying you can make in one move to reach the goal).

Heuristic 2 is admissible. This heuristic works in the same way as heuristic 1 except it divides the estimate by 3. This is still admissible, because heuristic 1 is, at all times, less than or equal to the solution cost, and heuristic 2 is underestimating heuristic 1.

Q3.2.B) A* Heuristic 1: (3,3,0,0,L) (3,1,0,2,R) (3,2,0,1,L) (3,0,0,3,R) (3,1,0,2,L) (1,1,2,2,R) (2,2,1,1,L) (0,2,3,1,R) (0,3,3,0,L) (0,1,3,2,R) (0,2,3,1,L) (0,0,3,3,R)

A* Heuristic 2: (3,3,0,0,L) (3,1,0,2,R) (3,2,0,1,L) (3,0,0,3,R) (3,1,0,2,L) (1,1,2,2,R) (2,2,1,1,L) (0,2,3,1,R) (0,3,3,0,L) (0,1,3,2,R) (0,2,3,1,L) (0,0,3,3,R)

Heuristic	Solution Path	Total Cost	Node Expansions	Same cost as UCS
h1	See Above	23	14	Yes
h2	See Above	23	15	Yes
h3	See Below	23	14	Yes

The more efficient heuristic was heuristic 1. This is because heuristic 1 was closer to the true cost at each step in its estimate. This led to more accurate predictions and resulted in less nodes expanded. Heuristic 2 is much more of an underestimate which leads to less accurate predictions.

Q3.2.C) A* Heuristic 3: (3,3,0,0,L) (3,1,0,2,R) (3,2,0,1,L) (3,0,0,3,R) (3,1,0,2,L) (1,1,2,2,R) (2,2,1,1,L) (0,2,3,1,R) (0,3,3,0,L) (0,1,3,2,R) (0,2,3,1,L) (0,0,3,3,R)

$$h(s) = X + \lceil X/3 \rceil - Y$$

$$X = 2 * MI + CI$$

$$Y = 1 \text{ if boat on left and } X > 0, \text{ else } Y = 0$$

MI = missionaries on the left

CI = cannibals on the left

This heuristic is admissible, because it assumes that the boat takes the most weight possible on each trip (3) to the right bank and the least weight possible (1) on each trip to the left. This ignores all restrictions on the amount of cannibals versus missionaries on each bank, making the heuristic underestimate the cost to the solution at every step, never over estimating. This dominates heuristic 1, because it is even closer to the actual cost of the solution at every step without actually surpassing, making its estimates more accurate.